

Risk Assessment and Mitigation Report

Moore's Comics and Pizza Palace POS System

Overview

For years now, Moore's Comics and Pizza Palace has been devoted to meeting the needs of consumers that come here to eat pizza, buy comics, and enjoy our other merchandise. Many enjoy coming to our lair; so many in fact, that we can no longer keep up with our sales by hand. We've even tried hiring a math major from college for an "internship" but he wants more money that we simply don't know if we have. So this is why we have decided to install our own point-of-sales system to keep track of our sales and budget and keep our business, hopefully booming, but at the very least afloat.

Our POS system's daily functions are including but not limited to:

- Conducting monetary transactions generated by cash and card
- Recording employee clock-in clock-out times to sum their paychecks
- Adding what-if scenarios to determine if we can afford new utilities
- Keeping track of our new utilities/taxes and how much everything costs
- Printing our total budget; in either profits or debts

Our POS system is connected to our cash registers and our iPad that is used to keep track of our employee times, as well as an app on our work computer that automatically sums up the calculations of what we make and what we owe. This allows us to make a plan to pay off our debt or in the event of a major profit, to put in a new asset to make our business bigger.

Risk Vulnerability

Stakeholders and Assets

Our assets include our technology that we use to perform the functions above. If anything goes wrong, our stakeholders include our business partners, employees, and our loyal consumers that want to hang out here after school. If our information is lost, stolen, or corrupted, we lose our business and can't pay our employees. On top of that, if we have to shut down, people can't come here to hang out and browse our merchandise while enjoying our food. We might be able to get by without buying any new technology hookups but interest in our store varies from week to week.

Risks

- Password security
- Theft
- Natural disasters

A major risk I have identified is password security. Employees are given a four-digit number, all of which start with either zero or one. Managers have a five-digit number that start with 21. Anyone could break into the computer system holding our money so lack of password security could lead to theft. Natural disasters that include wind, rain, sleet, and snow could very easily result in a power outage and shut down our system holding the funds. In New England, these storms occur at least once a year.

Valuation Matrix Definitions

Likelihood	Definition
High (3)	This threat is very likely to occur with little to no system in place.
Medium (2)	This threat is less likely to occur but it still couldn't hurt to add some system of protection.
Low (1)	There is little to no reason to believe this threat will occur.

Impact	Definition
High (3)	If this threat occurs, we will lose access to some very important information, as well as compromise our company's well-being and reputation.
Medium (2)	If this threat occurs, we will lose some important information and money but not enough to deal with any long-term ramifications.
Low (1)	If this threat occurs, we have a system to ensure no other problems.

Threat	Likelihood	Impact	Risk
Password security	Medium (2)	High (3)	Medium (6)
Theft	Low (1)	High (3)	Low (3)
Natural disasters	High (3)	Medium (2)	Medium (6)

Mitigation

We have just determined that our two biggest risks are password security and natural disasters. Theft is another one but lack of password security is what leads to that so let's start off with password security.

To implement fingerprint security in code, we need to leverage our devices' built-in biometric authentication through the operating system's application programming interfaces (API), checking if our device supports fingerprint scanning, prompting the user to authenticate with their fingerprint, and verifying the result with respect to our stored fingerprint data. For this, we want a secure keystone to store the encrypted fingerprint information. There is a `FingerprintManager` class in JavaScript that can be used to perform this operation. If we use fingerprint security, that should ensure that only we in any circumstances have access to our device information, thus mitigating theft as well.

Our other risk to mitigate is natural disasters, particularly that they could shut down our electricity system. One of the best ways to mitigate this is to install a generator. In the event of a power outage, a generator will make its own electricity for the store, thus protecting our POS system from natural disasters and saving information on our budget. Lowe's carries some really great generators; 10,100 watts and up will power the whole store.

WORKS CITED

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